Short-Horizon PM2.5 Forecasting in Beijing via Gradient Boosting with Rolling Chronological Backtesting

Manan Gilhotra, Harshita, Satyam, Yashi Agarwal, Akhil Mishra
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Code: https://github.com/MananGilhotra/air-quality-forecasting

Abstract—Fine particulate matter (PM2.5) is one of the most harmful urban air pollutants, and a reliable short-term forecast allows people and agencies to act before pollution peaks. This work forecasts hourly PM2.5 concentration in Beijing at a fixed horizon of six hours using the UCI Beijing PM2.5 dataset (43,824 hourly records, 2010–2014). We engineer 31 features from lagged pollution, rolling means, calendar terms and weather covariates, and train an XGBoost regressor. The model is compared with persistence and seasonal naive baselines under a four-fold rolling chronological backtest that never trains on future data. XGBoost achieves an average mean absolute error (MAE) of 35.76 μg/m^3 against 40.73 μg/m^3 for persistence, a 12% reduction, raises R^2 from 0.518 to 0.652, and wins in every test period. Forecasts are also mapped to US EPA (2024) Air Quality Index categories, where XGBoost assigns the correct category for 53.9% of hours.

Index Terms—PM2.5 forecasting, air quality, gradient boosting, XGBoost, time series, rolling-origin evaluation, feature engineering

I. Introduction

PM2.5 particles are small enough to penetrate deep into the lungs and are associated with cardiovascular and respiratory disease. Beijing has historically experienced severe PM2.5 episodes, particularly during the winter heating season [1]. A forecast several hours ahead gives residents time to limit outdoor exposure and gives authorities time to issue warnings.

The objective of this project is to forecast hourly PM2.5 at a fixed six-hour horizon and to test whether a machine learning model genuinely outperforms simple forecasting rules when evaluated only on future data. Manually chosen lag windows may not capture longer temporal dependencies or non-linear pollutant–weather interactions; gradient-boosted trees address the second issue, while sequence models such as long short-term memory (LSTM) networks [9] are a natural next step for the first.

The main contributions are: (i) a leakage-safe cleaning and feature pipeline for the UCI Beijing PM2.5 data; (ii) a rolling chronological backtest comparing XGBoost with persistence and seasonal naive baselines; and (iii) a mapping of forecasts to the US EPA 2024 AQI categories, all released as a reproducible code repository.

II. Literature Review

A. Dataset and Pollution Drivers

Liang et al. [1] introduced the hourly Beijing PM2.5 and meteorological records that underlie the UCI dataset [2]. Using two quasi-experiments, the 2014 APEC meeting and the annual winter heating, they showed that emissions and weather jointly drive PM2.5 and identified the heating season as a major contributor. These findings motivate our calendar and weather features.

B. Machine Learning and Tree Ensembles

A systematic review by Rybarczyk and Zalakeviciute [3] found that deterministic models struggle with the non-linear relationship between pollutant concentrations and their emission and dispersion sources, and that machine learning approaches are a promising alternative. XGBoost [4] has been applied directly to this problem: Pan [5] used it to predict hourly PM2.5 in Tianjin and reported that it outperformed random forest, multiple linear regression, decision tree and support vector regression. Zhai and Chen [6] built a stacked ensemble for daily Beijing PM2.5 that combined LASSO, AdaBoost, XGBoost and a genetic-algorithm-optimised multilayer perceptron through support vector regression, guided by exploratory analysis and feature selection.

C. Deep Sequence Models

Li et al. [7] developed an LSTM-based model that exploits spatiotemporal correlations and auxiliary data for hourly air pollutant prediction and showed improvements over several statistical and machine learning baselines. Huang and Kuo [8] combined convolutional and LSTM layers (APNet) on the same Beijing dataset used here, evaluating with MAE, RMSE, Pearson correlation and index of agreement.

D. Forecast Evaluation Practice

Tashman [10] formalised out-of-sample tests, including rolling-origin evaluation, in which the forecast origin is moved forward repeatedly. Bergmeir and Benítez [11] studied cross-validation for time series and recommended blocked, order-preserving schemes. Hyndman and Athanasopoulos [12] recommend simple benchmarks such as

TABLE I
Summary of Reviewed Literature

Study	Data / Task	Method	Relevance to this work
Liang et al. [1]	Beijing PM2.5 and weather, 2010–2015	Statistical analysis of quasi-experiments (APEC, winter heating)	Source of the dataset; motivates weather and seasonal features
Rybarczyk & Zalakeviciute [3]	Review of 46 air quality studies	Systematic review of ML approaches	Supports ML for non-linear pollution modelling
Pan [5]	Hourly PM2.5, Tianjin	XGBoost vs. RF, MLR, DT, SVR	XGBoost outperformed other methods for hourly PM2.5
Zhai & Chen [6]	Daily PM2.5, Beijing	Stacked ensemble (LASSO, AdaBoost, XGBoost, GA-MLP; SVR meta-learner)	EDA-driven feature engineering with tree models
Li et al. [7]	Hourly air pollutants	LSTM with spatiotemporal and auxiliary data	Basis for the planned LSTM extension
Huang & Kuo [8]	UCI Beijing PM2.5 (same data)	CNN-LSTM (APNet)	Deep learning reference on identical data
Tashman [10]; Bergmeir & Benítez [11]	Forecast evaluation methodology	Rolling-origin and blocked validation	Justifies our rolling chronological backtest

the naive (persistence) and seasonal naive methods as reference points for any forecasting model.

E. Research Gap and Positioning

Table I summarises the reviewed work. Rather than proposing a new architecture, this project focuses on a fair and reproducible comparison: a gradient boosting model is evaluated against explicit persistence and seasonal naive benchmarks [12] under a rolling-origin backtest [10], with leakage controls and a stated AQI standard [13]. Sequence models [7], [8] are left as the next step.

III. Dataset and Exploratory Analysis

A. Dataset

The UCI Beijing PM2.5 dataset [1], [2] contains hourly readings from the US Embassy in Beijing from 1 January 2010 to 31 December 2014 (Table II). Only PM2.5 has missing values: 2,067 hours (4.7%), mostly in 2010–2012, with a longest gap of 155 hours.

TABLE II
Dataset Variables

Column	Meaning	Unit
pm2.5	PM2.5 concentration (target)	μg/m^3
DEWP	Dew point	°C
TEMP	Temperature	°C
PRES	Air pressure	hPa
cbwd	Wind direction (NE, NW, SE, calm)	category
Iws	Cumulated wind speed	m/s
Is, Ir	Cumulated hours of snow, rain	h

B. Exploratory Findings

PM2.5 is strongly right-skewed (median 72, mean 98.6, maximum 994 μg/m^3), so rare severe episodes inflate RMSE relative to MAE. Fig. 1 shows clear daily and yearly cycles: concentrations peak near midnight (about 113 μg/m^3) and fall to about 86 μg/m^3 in mid-afternoon, and they are highest in the heating months (February about 126 μg/m^3). Wind direction is the strongest weather signal: median PM2.5 is 31 μg/m^3 under north-westerly wind and 98 μg/m^3 in calm conditions. Linear correlations with weather are weak (wind speed -0.25, dew point 0.17), suggesting non-linear effects. Finally, PM2.5 is highly persistent (Fig. 2): its autocorrelation is 0.97 at 1 h, 0.75 at 6 h and 0.40 at 24 h, fading after about two days.

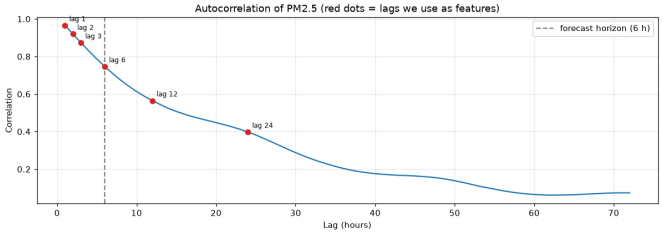


Fig. 2. Autocorrelation of PM2.5 for lags of 1–72 hours. Red points mark the lags used as features; the dashed line marks the 6-hour horizon.

IV. Methodology

A. Cleaning

Records are sorted by time and resampled to a strict hourly grid so that a shift of six rows always equals six hours. A binary flag pm25_missing is added before any filling. Gaps are then forward filled using past values only; interpolation is avoided because it would use a later observation and leak future information. The fully missing first day is dropped and wind direction is one-hot encoded, giving 43,800 hourly rows with 2,043 flagged hours.

B. Target and Features

The target is PM2.5 exactly six hours after the forecast time. A companion flag records whether that future value

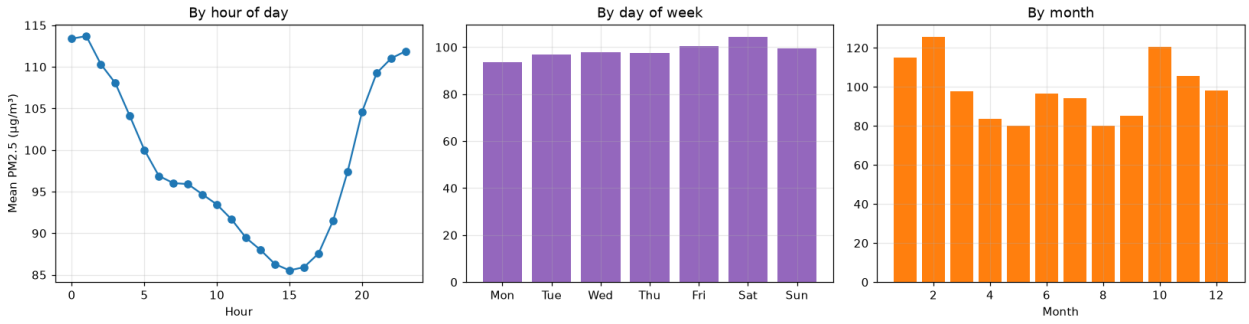


Fig. 1. Mean PM2.5 by hour of day, day of week and month (2010–2014).

was genuinely measured; only such rows are used for training and scoring. Thirty-one features are built from current and past hours only (Table III). Hour and month are additionally encoded as $\sin(2\pi h/24)$, $\cos(2\pi h/24)$ and the monthly equivalents so that adjacent times remain close. The seasonal reference value is PM2.5 at the same clock hour one day before the target, i.e., a shift of $24 - 6 = 18$ hours. No feature scaling is applied, as tree models are invariant to monotonic rescaling.

TABLE III
Engineered Features (31 in Total)

Group	No.	Contents
Current PM2.5	1	PM2.5 at forecast time
Lags	7	1, 2, 3, 6, 12, 24 h; same hour yesterday
Rolling means	2	Means over last 6 h and 24 h
Calendar	8	Hour, weekday, month, weekend, sin/cos of hour and month
Weather	8	Six raw covariates, temperature–dew point gap, 3-h pressure change
Wind direction	4	One-hot NE, NW, SE, calm
Missing flag	1	PM2.5 measured this hour

C. Models

Persistence forecasts the current value: $\hat{y}_{t+6} = y_t$. Seasonal naive forecasts $\hat{y}_{t+6} = y_{t-18}$. XGBoost [4] fits 300 gradient-boosted regression trees (learning rate 0.05, maximum depth 6, 80% row and column subsampling, fixed random seed); each tree is trained on the residuals of the ensemble so far.

D. Rolling Chronological Backtest

Following rolling-origin evaluation [10], [11], each fold trains on all data before its test period and tests on the next six months (Fig. 4); the four test periods cover January 2013 to December 2014, about 4,300 measured hours each. As a leakage guard, the last six training hours, whose targets fall inside the test period, are kept as history but excluded as training targets. All models are scored on the same measured hours using

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \quad \mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \quad (1)$$

and $R^2 = 1 - \sum(y_i - \hat{y}_i)^2 / \sum(y_i - \bar{y})^2$, averaged over the four folds.

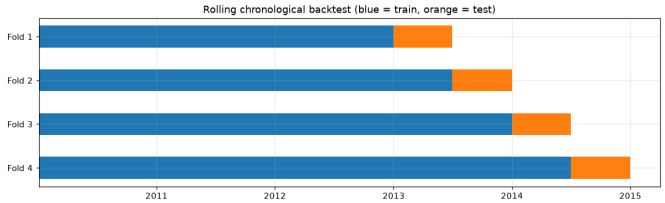


Fig. 4. Rolling chronological backtest: expanding training window (blue) and six-month test periods (orange).

E. AQI Mapping

Forecasts are labelled with the US EPA 2024 PM2.5 AQI categories [13]: Good ($\leq$9.0), Moderate ($\leq$35.4), Unhealthy for Sensitive Groups ($\leq$55.4), Unhealthy ($\leq$125.4), Very Unhealthy ($\leq$225.4) and Hazardous (above 225.4 μg/m^3). Because EPA defines these breakpoints on 24-hour averages, applying them to hourly forecasts is a stated simplification. AQI category accuracy is the share of hours whose forecast falls in the same category as the observation.

V. Results

Table IV reports the average over the four folds. XGBoost has the lowest error on every metric, reducing MAE by 12% relative to persistence. Fig. 3 shows that it also wins in each individual test period, by 8% to 16%. Seasonal naive performs worst, with negative R^2, because yesterday's level is a poor guide six hours ahead when pollution episodes build and clear within one or two days.

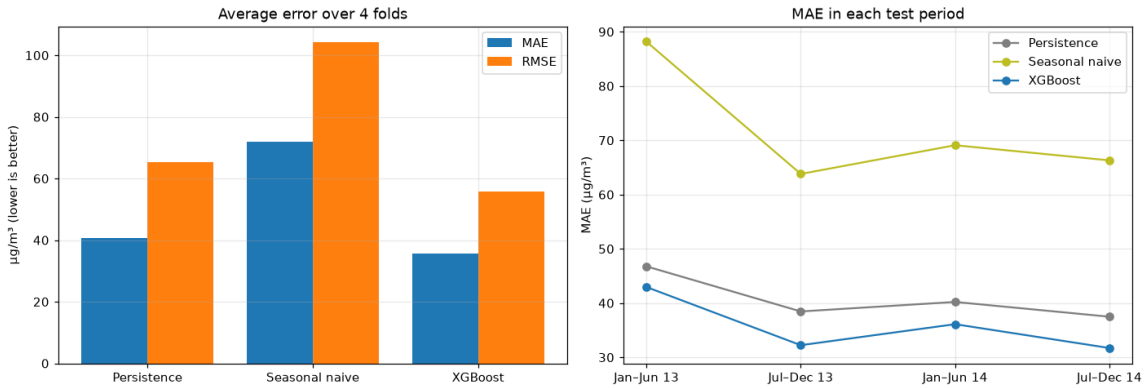


Fig. 3. Average MAE and RMSE over four folds (left) and MAE in each test period (right).

TABLE IV
Average Results over Four Folds (6-Hour Horizon)

Model	MAE	RMSE	R^2	AQI acc.
Persistence	40.73	65.31	0.518	49.8%
Seasonal naive	71.84	104.19	−0.223	31.5%
XGBoost	35.76	55.80	0.652	53.9%

MAE and RMSE in $\mu g/m^3$.

Fig. 5 shows the final week of the test data. Persistence reproduces the observed curve six hours late by construction, whereas XGBoost often turns earlier. Errors are largest in January and February (MAE about 62 and 50 $\mu g/m^3$), during heating-season spikes that have no precursor in the recent history, and smallest in summer (about 24–27 $\mu g/m^3$).

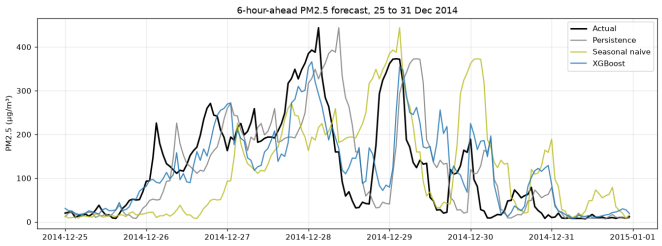


Fig. 5. Six-hour-ahead forecasts versus observed PM2.5, 25–31 December 2014.

Feature importance from the final-fold model (Fig. 6) attributes about 30% to current PM2.5, consistent with the strong persistence seen in Fig. 2, followed by calendar features (20%), wind direction (16%) and weather (16%). The hand-picked lags carry only 12%, and the single most important weather column is north-westerly wind.

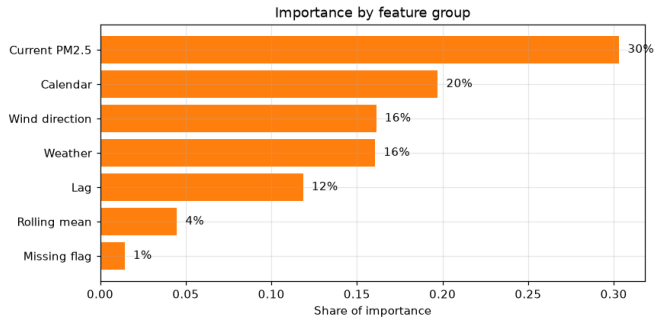


Fig. 6. XGBoost feature importance by feature group (final fold).

VI. Discussion and Limitations

Persistence captures most short-term signal because PM2.5 changes slowly; XGBoost adds value by learning when concentrations are likely to rise or fall, for example under calm air at night in winter or before a north-westerly wind clears the air. The main limitations are: (i) sudden spikes remain hard to anticipate from six hours of history; (ii) lag windows are fixed and end at 24 h; (iii) results cover a single station from 2010–2014; (iv) 24-hour AQI breakpoints are applied to hourly values; and (v) only observed, not forecast, weather is used.

VII. Conclusion and Future Work

Under a rolling chronological backtest, an XGBoost model with lag, rolling, calendar and weather features forecasts six-hour-ahead PM2.5 in Beijing with 12% lower MAE than persistence (35.76 vs. 40.73 $\mu g/m^3$) and wins in all four test periods. The pipeline is fully reproducible, with notebooks, figures and automated tests in the project repository. Future work will train an LSTM [7], [9] on raw 48-hour sequences, add weather forecasts and neighbouring stations, tune hyperparameters on a time-ordered validation split and forecast 24-hour averages for a fully standard AQI.

Team Contributions

Manan Gilhotra: project setup, data loading and cleaning. Harshita: feature engineering and target. Satyam: rolling backtest, metrics and baselines. Yashi Agarwal: XGBoost model. Akhil Mishra: AQI mapping, charts, pipeline runner and dashboard.

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